

Finite-Size Crossover in Ollivier-Ricci Curvature of Random Regular Graphs: Exact Computation via Linear Programming

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February 2026

Abstract

We confirm a sign change in mean Ollivier-Ricci curvature $\bar{\kappa}$ of random k -regular graphs at a critical density η_c , computed via exact linear programming (no entropy regularization). For $N = 100$ vertices averaged over 10 random seeds, the transition occurs between $\eta = 1.96$ ($\bar{\kappa} = -0.016$, $p < 10^{-6}$) and $\eta = 2.56$ ($\bar{\kappa} = +0.022$, $p < 10^{-6}$), with the sign change confirmed by one-sample t -tests. Multi- N scaling ($N \in \{50, 100, 200, 500, 1000\}$) yields finite-size scaling $\eta_c(N) = \eta_c^\infty - a/\sqrt{N}$ with $\eta_c^\infty \approx 3.7$ (fixed $\beta = 1/2$, $R^2 = 0.995$; free- β fit gives $\eta_c^\infty \approx 4.2$ with wider uncertainty). A free-exponent fit gives $\beta = 0.35$ (profile 95% CI $[0.20, 0.53]$, containing $\beta = 0.5$); the transition slope at η_c scales as $N^{-0.20}$, consistent with a crossover rather than a sharp phase transition in the finite-size regime studied. Comparison with Sinkhorn approximation ($\varepsilon = 0.01$) confirms mean bias $|\Delta\kappa| < 0.015$ and identical sign-change location, validating Sinkhorn for practical use. Lin-Lu-Yau curvature (without idleness) shows the same monotonic trend but a different sign-change location, consistent with the dependence on the random walk parameter α . Erdős-Rényi graphs $G(N, p)$ with matched expected degree also exhibit a sign change, but at lower critical density ($\eta_c^{\text{ER}} \approx 1.9$ vs. $\eta_c^{\text{reg}} \approx 2.3$ at $N = 100$), demonstrating that the transition is robust across graph models while the critical point depends on degree distribution. A Lean 4 formalization (29 modules, 13 explicit axioms, with 0 sorry in the 7 core ORC-theory modules) provides machine-checked proofs of Wasserstein non-negativity, coupling marginals, probability measure normalization, curvature bounds, regime exclusivity, and clustering bounds. The full analytical characterization of the sign change remains open.

Keywords: Ollivier-Ricci curvature; finite-size crossover; sign change; optimal transport; linear programming; random regular graphs; Erdős-Rényi graphs; Lean 4; formal verification

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1 Introduction

1.1 Motivation

Ollivier-Ricci curvature [1] generalizes Ricci curvature to discrete metric spaces via optimal transport. For an edge (u, v) in a graph G :

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)}, \quad (1)$$

where $\mu_u = \alpha\delta_u + (1 - \alpha) \cdot \text{Uniform}(N(u))$ and W_1 is the Wasserstein-1 distance. The sign of κ classifies local geometry: $\kappa < 0$ (hyperbolic/tree-like), $\kappa = 0$ (flat), $\kappa > 0$ (spherical/clique-like).

Network curvature has found broad application: community detection [2], cancer network differentiation [10], and bottleneck identification in graph neural networks [12]. Empirical studies reveal that many real-world networks—particularly semantic networks and biological connectomes—are hyperbolic ($\bar{\kappa} < 0$), with the density parameter $\eta = k^2/N$ governing the geometric regime [2, 11]. This raises the question: is there a sign change in $\bar{\kappa}$ as η crosses a critical value, and is it a sharp transition or a crossover?

1.2 Contributions

1. **Exact computation:** We compute Wasserstein-1 distances via linear programming (JuMP + HiGHS), eliminating the entropy regularization bias of Sinkhorn approximation. With 10 random seeds for $N = 100$, we confirm the sign change at $\eta_c \approx 2.2$ with 95% confidence intervals and statistical hypothesis tests.
2. **Method comparison:** We quantify the Sinkhorn bias: mean $\Delta\kappa = -0.003$, maximum $|\Delta\kappa| = 0.014$. Both methods agree on the sign-change location ($k = 16$, $\eta = 2.56$). We also compare against Lin-Lu-Yau curvature (without idleness), which shows the same monotonic trend but a shifted sign-change location, and against Erdős-Rényi $G(N, p)$ graphs, which exhibit the same sign change at a lower critical density.
3. **Multi- N scaling and critical exponents:** Exact LP sweeps across five system sizes ($N = 50$ –1000) reveal finite-size scaling

$$\eta_c(N) = \eta_c^\infty - a/\sqrt{N}, \quad \eta_c^\infty \approx 3.7, \quad R^2 = 0.995.$$

A free-exponent fit yields $\beta = 0.35$ (profile 95% CI $[0.20, 0.53]$), consistent with but not uniquely selecting $\beta = 1/2$. The transition slope at η_c scales as $N^{-0.20}$, indicating crossover behavior.

4. **Lean 4 formalization:** 29 modules, 13 explicit axioms (3 in core ORC modules, 2 in dynamic-network extensions, 8 in hypercomplex algebra; all standard), with **0 sorry** in the 7 core ORC-theory modules.

1.3 Limitations (stated upfront)

- The 7 core ORC-theory modules contain **0 sorry statements** and rely on **3 explicit axioms** in the core (adjacency coupling bound $W_1 \leq 2$, support-distance bound for the idleness measures, and specification bridge; all mathematically standard — Wasserstein symmetry, the triangle inequality, and the local clustering bound, previously axiomatized, are now machine-checked). Eight non-core modules carry 65 **sorry**-carrying declarations for ongoing formalization of spectral geometry, Ricci flow, random graphs, and related infrastructure. The sign change is formally *stated* but not formally *proven*.
- Remaining core axioms are the adjacency coupling bound (`wasserstein_le_two_of_adjacent`), the support-distance bound (`probabilityMeasure_support_dist_bound`), and the specification bridge. Wasserstein symmetry and the triangle inequality are machine-checked in `WassersteinProven.lean`; the local clustering bound is machine-checked in `Basic.lean`; probability measure normalization in `Curvature.lean`.
- The exact analytical form of $\mathbb{E}[\kappa]$ as a function of η remains open. The heuristic $(\eta - \eta_c)/(\eta + 1)$ does not fit the data ($R^2 < 0$).

2 Preliminaries

2.1 Wasserstein-1 Distance

For probability measures μ, ν on a finite set V with metric d :

$$W_1(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \sum_{x, y} d(x, y) \cdot \gamma(x, y), \quad (2)$$

where $\Gamma(\mu, \nu)$ is the set of couplings (joint distributions with marginals μ and ν). For finite graphs, this is a linear program:

$$\min \sum_{i, j} d_{ij} \gamma_{ij} \quad \text{s.t.} \quad \sum_j \gamma_{ij} = \mu_i, \quad \sum_i \gamma_{ij} = \nu_j, \quad \gamma_{ij} \geq 0. \quad (3)$$

2.2 Sinkhorn Approximation

Sinkhorn’s algorithm [3] solves an entropy-regularized version:

$$W_1^\varepsilon(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \sum_{x, y} d(x, y) \cdot \gamma(x, y) + \varepsilon \cdot H(\gamma), \quad (4)$$

where $H(\gamma) = -\sum \gamma_{ij} \log \gamma_{ij}$. As $\varepsilon \rightarrow 0$, $W_1^\varepsilon \rightarrow W_1$. We use $\varepsilon = 0.01$ throughout.

2.3 Lin-Lu-Yau Curvature

Lin, Lu, and Yau [6] defined a graph curvature without solving an optimal transport problem. For an edge (u, v) in a graph where the random walk has no idleness ($\alpha = 0$):

$$\kappa_{\text{LLY}}(u, v) = \frac{2|\Delta(u, v)| + 2 - \max(\deg u, \deg v)}{\max(\deg u, \deg v)}, \quad (5)$$

where $|\Delta(u, v)|$ is the number of common neighbors. For k -regular graphs: $\kappa_{\text{LLY}} = (2|\Delta(u, v)| + 2 - k)/k$. The key difference from ORC is the absence of idleness: LLY uses $\alpha = 0$ while our ORC computation uses $\alpha = 0.5$.

2.4 Random Regular Graphs

We use the configuration model (via Julia’s `Graphs.random_regular_graph`) to generate random k -regular graphs on N vertices. We vary k to sweep the density parameter $\eta = k^2/N$. For each (k, N) pair, we average over multiple random seeds (10 for $N = 100$, 5 for multi- N) to obtain ensemble statistics with 95% confidence intervals via the t -distribution.

3 Exact Computation of the Curvature Sign Change

3.1 Implementation

We implement exact Wasserstein-1 computation in Julia 1.12.4 using JuMP v1.29.4 [4] with the HiGHS v1.21.1 solver:

1. For each edge (u, v) : construct probability measures μ_u, μ_v with idleness $\alpha = 0.5$
2. Compute all-pairs shortest paths via BFS
3. Solve the LP for $W_1(\mu_u, \mu_v)$
4. Compute $\kappa(u, v) = 1 - W_1/d(u, v)$
5. Report mean, median, std over all edges

Computational complexity. Each LP has N^2 variables and $2N$ equality constraints, solvable in $O(N^3)$ time by interior-point methods. With $|E| = kN/2$ edges, the per-graph cost is $O(kN^4)$. Sinkhorn runs in $O(N^2/\varepsilon)$ per edge with much smaller constants, making it $\sim 100\times$ faster in practice.

3.2 Results: Sign Change at $N = 100$

The sign change occurs between $\eta = 1.96$ ($\bar{\kappa} = -0.016$) and $\eta = 2.56$ ($\bar{\kappa} = +0.022$). Linear interpolation gives $\eta_c \approx 2.2$. The 95% confidence intervals at both transition points exclude zero, confirming the sign change is not a statistical artifact. See Figure 2 for the complete sign-change curve.

Main Computational Result. *For random k -regular graphs on N vertices generated by the configuration model, the mean Ollivier-Ricci curvature $\bar{\kappa}(\eta)$ (with $\alpha = 0.5$, computed via exact LP) undergoes a sign change at a critical density $\eta_c(N)$ that satisfies finite-size scaling $\eta_c(N) = \eta_c^\infty - a/\sqrt{N}$ with $\eta_c^\infty \approx 3.7$ (confirmed across $N \in \{50, 100, 200, 500, 1000\}$). At $N = 100$, the transition is between $\eta = 1.96$ and $\eta = 2.56$ with $p < 10^{-6}$ at both boundaries.*

Table 1: Exact LP curvature for random k -regular graphs ($N = 100$, averaged over 10 seeds). 95% confidence intervals computed via the t -distribution with 9 degrees of freedom.

k	$\eta = k^2/N$	$\bar{\kappa}_{\text{exact}}$	95% CI	σ_{edges}	Geometry
2	0.04	+0.000	$[\pm 0.000]$	0.000	Flat
3	0.09	-0.277	$[-0.291, -0.263]$	0.164	Hyperbolic
4	0.16	-0.363	$[-0.377, -0.350]$	0.167	Hyperbolic
6	0.36	-0.302	$[-0.309, -0.294]$	0.127	Hyperbolic
8	0.64	-0.190	$[-0.196, -0.183]$	0.089	Hyperbolic
10	1.00	-0.114	$[-0.118, -0.109]$	0.074	Hyperbolic
12	1.44	-0.060	$[-0.063, -0.057]$	0.066	Hyperbolic
14	1.96	-0.016	$[-0.018, -0.013]$	0.060	Transition
16	2.56	+0.022	$[+0.019, +0.024]$	0.054	Transition
18	3.24	+0.056	$[+0.053, +0.059]$	0.047	Spherical
20	4.00	+0.084	$[+0.081, +0.086]$	0.041	Spherical
25	6.25	+0.127	$[+0.125, +0.130]$	0.036	Spherical
30	9.00	+0.151	$[+0.150, +0.152]$	0.034	Spherical
35	12.25	+0.166	$[+0.165, +0.167]$	0.033	Spherical
40	16.00	+0.179	$[+0.178, +0.181]$	0.032	Spherical

Statistical confirmation. One-sample t -tests for $H_0: \bar{\kappa} = 0$ at the transition points: at $k = 14$, $t = -14.72$ ($p = 1.3 \times 10^{-7}$); at $k = 16$, $t = +18.04$ ($p = 2.2 \times 10^{-8}$). Both strongly reject H_0 .

3.3 Sinkhorn vs. Exact Comparison

Summary statistics: mean bias = -0.003 , std = 0.006 , max |bias| = 0.014 . Both methods detect the sign change at $k = 16$ ($\eta = 2.56$) (Figure 3).

3.4 Qualitative Structure

Several patterns emerge from the exact results:

1. **Non-monotonic curvature:** $\bar{\kappa}$ first *decreases* from $k = 2$ to $k = 4$ (reaching -0.363), then monotonically increases. A 2-regular graph is a disjoint union of cycles; by symmetry of the shortest-path metric, $W_1 = 0$ and $\kappa = 0$.
2. **Wide per-edge variance at low k :** $\sigma_{\text{edges}} = 0.164$ at $k = 3$ versus 0.032 at $k = 40$.
3. **Asymptotic saturation:** $\bar{\kappa}$ approaches but does not reach $+1$ even at $k = 40$, consistent with the upper bound $\bar{\kappa} \leq 1$ proven formally.
4. **LLY comparison:** See Table 3.

Table 2: Per- k comparison of Sinkhorn ($\varepsilon = 0.01$) and exact LP curvature (single seed $s = 42$).

k	η	κ_{Sinkhorn}	κ_{exact}	$\Delta\kappa$
3	0.09	-0.252	-0.264	-0.011
4	0.16	-0.344	-0.353	-0.009
6	0.36	-0.303	-0.297	+0.005
8	0.64	-0.178	-0.192	-0.014
10	1.00	-0.122	-0.115	+0.007
12	1.44	-0.061	-0.059	+0.001
14	1.96	-0.015	-0.017	-0.002
16	2.56	+0.026	+0.022	-0.004
20	4.00	+0.089	+0.084	-0.005
30	9.00	+0.152	+0.151	-0.002
40	16.00	+0.181	+0.181	-0.000

Table 3: ORC vs. LLY curvature ($N = 100$, 10 seeds). The difference peaks at $k = 20$ ($\eta = 4.0$).

k	η	$\bar{\kappa}_{\text{ORC}}$	$\bar{\kappa}_{\text{LLY}}$	$ \text{ORC} - \text{LLY} $
3	0.09	-0.277	-0.313	0.036
6	0.36	-0.302	-0.595	0.293
10	1.00	-0.114	-0.660	0.546
14	1.96	-0.016	-0.643	0.627
20	4.00	+0.084	-0.587	0.671
30	9.00	+0.151	-0.462	0.613
40	16.00	+0.179	-0.341	0.520

LLY is systematically more negative than ORC, and **LLY does not cross zero** in the tested range ($\eta \leq 16$). This is expected: idleness $\alpha = 0.5$ reduces transport cost and shifts the sign change to lower η .

3.5 Erdős-Rényi Comparison

Key observations:

1. **ER graphs also exhibit a sign change** (Figure 5), with $\eta_c^{\text{ER}} \approx 1.90$.
2. **ER transitions earlier:** $\eta_c^{\text{ER}} \approx 1.90$ versus $\eta_c^{\text{reg}} \approx 2.26$, a difference of ≈ 0.36 , consistent with the asymptotic analysis of Mitsche and Mubayi [16] for $G(n, p)$.
3. **ER is systematically more positive:** $\Delta > 0$ at every $k \geq 3$.
4. **Higher variance in ER:** $\sigma_{\text{ER}} \in [0.002, 0.034]$ versus $\sigma_{\text{reg}} \in [0.001, 0.005]$.

Table 4: Erdős-Rényi $G(100, p)$ curvature vs. random k -regular graphs ($N = 100$, 10 seeds each).

k	η	$\bar{\kappa}_{\text{regular}}$	$\bar{\kappa}_{\text{ER}}$	σ_{ER}	Δ (ER-reg)
3	0.09	-0.277	-0.238	0.018	+0.039
6	0.36	-0.302	-0.247	0.025	+0.054
10	1.00	-0.114	-0.093	0.013	+0.020
12	1.44	-0.060	-0.040	0.008	+0.019
14	1.96	-0.016	+0.005	0.008	+0.021
16	2.56	+0.022	+0.046	0.008	+0.024
20	4.00	+0.084	+0.107	0.005	+0.024
30	9.00	+0.151	+0.166	0.003	+0.015
40	16.00	+0.179	+0.208	0.002	+0.029

3.6 Multi- N Scaling

Table 5: Critical point location as a function of N .

N	Seeds	Last $\bar{\kappa} < 0$	First $\bar{\kappa} > 0$	η_c (interpolated)
50	5	$k = 8$ ($\eta = 1.28$)	$k = 10$ ($\eta = 2.00$)	≈ 1.73
100	10	$k = 14$ ($\eta = 1.96$)	$k = 16$ ($\eta = 2.56$)	≈ 2.22
200	5	$k = 20$ ($\eta = 2.00$)	$k = 25$ ($\eta = 3.13$)	≈ 2.71
500	5	$k = 35$ ($\eta = 2.45$)	$k = 40$ ($\eta = 3.20$)	≈ 3.09
1000	3	$k = 56$ ($\eta = 3.14$)	$k = 58$ ($\eta = 3.36$)	≈ 3.32

Key observations (see Figures 2 and 4):

1. η_c **increases with** N : from ≈ 1.73 at $N = 50$ to ≈ 3.32 at $N = 1000$.
2. **Finite-size scaling**: $\eta_c(N) = \eta_c^\infty - a/\sqrt{N}$ with $\eta_c^\infty \approx 3.75$, $a \approx 14.62$, $R^2 = 0.995$.
3. **Transition sharpens with** N : At $N = 50$, the transition spans $\Delta\eta \approx 0.7$. At $N = 1000$, $\sigma < 0.0006$ and $\Delta\eta = 0.23$.
4. **Curvature magnitude increases at low** η : At $\eta = 0.16$, $\bar{\kappa} = -0.266$ ($N = 50$), -0.363 ($N = 100$), -0.421 ($N = 200$).

3.7 Critical Exponents and Crossover Character

Free-exponent scaling fit. Releasing the exponent to a free parameter, we fit $\eta_c(N) = \eta_c^\infty - a/N^\beta$. The best fit gives

$$\eta_c^\infty = 4.20, \quad a = 9.76, \quad \beta = 0.350 \quad (R^2 = 0.998), \quad (6)$$

improving slightly over the fixed- $\beta = 0.5$ fit ($R^2 = 0.995$). The profile 95% CI spans $\beta \in [0.195, 0.525]$; $\beta = 0.5$ lies within this interval. **Caution:** this fit uses 5 data points with 3 free parameters, leaving only 2 degrees of freedom. The 95% CI for η_c^∞ from the free fit is approximately $[3.2, 5.8]$ (profile likelihood), substantially wider than the fixed- β estimate of 3.75 ± 0.15 . We report the fixed- $\beta = 0.5$ value $\eta_c^\infty \approx 3.7$ as the primary estimate, noting that the free- β fit suggests the true asymptotic value may be higher.

Transition slope scaling. Log-log regression of the transition slope $\Delta\bar{\kappa}/\Delta\eta$ at η_c yields

$$\left. \frac{d\bar{\kappa}}{d\eta} \right|_{\eta_c(N)} \sim N^{-0.20}. \quad (7)$$

The **decreasing slope** is a crossover signature. Whether a sharp transition emerges for $N \gg 1000$ remains open.

Data collapse. Minimizing residual variance of a cubic polynomial fit in the rescaled variable $(\eta - \eta_c(N)) \cdot N^\gamma$ gives optimal $\gamma = 0.42$, reducing variance by 22% relative to mean-field $\gamma = 0.5$.

Physical interpretation: triangles per edge at the transition. For a k_c -regular random graph with $k_c = \sqrt{\eta_c(N) \cdot N}$:

$$C(N) \approx \frac{k_c}{N} = \sqrt{\frac{\eta_c(N)}{N}} \rightarrow 0 \quad \text{as } N \rightarrow \infty, \quad (8)$$

$$\langle \Delta \rangle_{\text{edge}}(N) \approx \frac{k_c^2}{N} = \eta_c(N) \rightarrow \eta_c^\infty \approx 3.7 \quad \text{as } N \rightarrow \infty. \quad (9)$$

The sign change in ORC is governed by the **absolute** number of triangles per edge, not by the local triangle fraction. Below η_c , mass transport requires long detours through tree-like neighborhoods ($W_1 > d$, $\kappa < 0$); above it, triangular shortcuts reduce W_1 below d ($\kappa > 0$).

4 Lean 4 Formalization

4.1 Architecture

The formalization uses Lean 4 (v4.17.0) with Mathlib (v4.17.0). It consists of 29 modules totaling 10 212 lines with **13 explicit axioms** (3 in the 7 core ORC-theory modules; 10 in extension modules for dynamic networks and hypercomplex algebra). The 7 core ORC-theory modules contain **0 sorry**; eight non-core modules carry 65 **sorry**-carrying declarations (**lake** build count, 2026-07-11); the remaining 14 extension modules are **sorry**-free.

4.2 Machine-Checked Results

The following are fully proven with no axioms:

Wasserstein distance.

- **Non-negativity:** $W_1(d, \mu, \nu) \geq 0$ when $d \geq 0$ (`Wasserstein.wasserstein_nonneg`)
- **Coupling marginal lemma:** If $\nu(y) = 0$ then $\gamma(x, y) = 0$ for any coupling with marginal ν (`Wasserstein.coupling_zero_of_marginal_zero`)
- **LP equivalence:** The LP formulation equals the abstract Wasserstein definition (`Wasserstein.lp_equals_wasserstein`)

Curvature.

- **Unreachable nodes:** $\kappa(u, v) = 0$ when u and v are in different connected components (`Curvature.curvature_no_path`)
- **Regime exclusivity:** Hyperbolic and spherical regimes are mutually exclusive (`Curvature.regimes_exclusive`)
- **Probability measure normalization:** $\sum_v \mu_u(v) = 1$ for the idleness-based measure at nodes with positive degree (`Curvature.probabilityMeasure_normalization_proven`)

4.3 Axiomatized Results

Core ORC-theory axioms (3): adjacency coupling bound ($W_1(\mu_u, \mu_v) \leq 2$ for adjacent u, v ; the $d = 1$ case behind $\kappa \in [-1, 1]$), support-distance bound ($d(x, y) \leq d(u, v) + 2$ on the supports of the idleness measures), and specification bridge (Julia implementation satisfies the formal ORC specification). Wasserstein symmetry, the triangle inequality, and the local clustering bound, previously axiomatized, are now machine-checked.

Extension module axioms (10): `DynamicNetworks.lean` (2 axioms): ORC Lipschitz continuity under edge perturbations; brain connectome sweet-spot hypothesis. `Hypercomplex.lean` (8 axioms): standard properties of octonion and sedenion algebra.

Note: Probability measure normalization, previously axiomatized, is now **fully machine-checked** in `Curvature.lean`.

4.4 Verbatim Proofs

Wasserstein non-negativity. The complete, machine-checked proof:

```
-- Wasserstein distance is non-negative when d is non-negative.
-/
lemma wasserstein_nonneg
  (hmu : ProbabilityMeasure.IsProbabilityMeasure mu)
  (hnu : ProbabilityMeasure.IsProbabilityMeasure nu)
  (h_nonneg : forall u v, 0 <= d u v) :
  0 <= wasserstein1 d mu nu := by
  apply Real.sInf_nonneg
  intro c hc
  rcases hc with <gamma, hgamma>
```

```

rw [<-hgamma]
simp [couplingCost]
apply Finset.sum_nonneg
intro u _
apply Finset.sum_nonneg
intro v _
apply mul_nonneg
. exact h_nonneg u v
. exact gamma.gamma_nonneg u v

```

Regime exclusivity.

```

theorem regimes_exclusive (alpha : Idleness) :
  Not (isHyperbolic G alpha /\ isSpherical G alpha) := by
  intro <h_hyp, h_sph>
  simp [isHyperbolic, isSpherical] at h_hyp h_sph
  linarith

```

5 Discussion

5.1 The Critical Point

Our exact LP computation places the critical point at $\eta_c \approx 2.2$ for $N = 100$. The multi- N scaling (§3) reveals that η_c increases with N : from ≈ 1.7 at $N = 50$ to ≈ 3.3 at $N = 1000$. The finite-size scaling fit $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ ($R^2 = 0.995$) extrapolates to $\eta_c^\infty \approx 3.7$ (Figure 4). The free-exponent fit (§3) gives $\eta_c^\infty = 4.20$ with $\beta = 0.35$; the discrepancy with 3.75 reflects the limited constraining power of 5 data points with 3 free parameters. The $N = 1000$ data point (4.5 hours, 24 threads, ~ 500 LP instances) confirms the fixed- β fit: predicted $\eta_c(1000) = 3.29$, actual $\eta_c(1000) = 3.32$ (residual +0.03).

Crossover character. Section 3 provides evidence that the transition, over $N \leq 1000$, is better characterized as a crossover than a sharp phase transition: the slope of $\bar{\kappa}$ at η_c decreases as $N^{-0.20}$. The free-exponent fit ($\beta = 0.35$, 95% CI [0.20, 0.53]) does not uniquely identify $\beta = 1/2$.

5.2 Method Comparison

Sinkhorn with $\varepsilon = 0.01$ is remarkably accurate: mean bias -0.003 , no shift in the detected critical point. The Lin-Lu-Yau curvature ($\alpha = 0$) is systematically more negative and does not change sign in the tested range, consistent with general theory.

5.3 Formalization: What Is Proven and What Is Not

Achieved. The 7 core ORC-theory modules compile with **0 sorry**. Key structural results ($W_1 \geq 0$, coupling marginal vanishing, probability measure normalization, curvature

vanishing for unreachable nodes, regime exclusivity, average clustering bounds) are fully machine-checked.

Not achieved. The sign change is not formally proven as a sharp transition. Main barriers: (1) random graph probability infrastructure in Lean 4 is immature; (2) expected curvature calculation requires intricate combinatorial arguments; (3) connecting computational results to formal specifications remains at the axiom level.

5.4 Open Problems

1. **Analytical critical point:** Derive a closed-form expression for $\eta_c(\alpha)$ as a function of idleness and graph model parameters. Hehl’s explicit ORC formulas for regular graphs [14], which reduce curvature to a min-cost assignment on exclusive neighborhoods, provide a potential starting point.
2. **α -dependence:** Sweep $\alpha \in [0, 1]$ to map the full (α, η_c) phase boundary.
3. **Scaling exponent and transition character:** A free-exponent fit gives $\beta = 0.35$ (profile 95% CI $[0.20, 0.53]$). Whether a true sharp transition emerges for $N \gg 1000$ requires larger- N simulations and analytic theory.
4. **Universality and metric-space robustness:** Our ER comparison shows the transition is robust but the critical point depends on degree distribution ($\eta_c^{\text{ER}} \approx 1.9$ vs. $\eta_c^{\text{reg}} \approx 2.3$ at $N = 100$).
See Appendix E for preliminary hypercomplex embedding results suggesting that the sign change depends on the transport metric.
5. **Functional form of $\mathbb{E}[\kappa(\eta)]$:** The heuristic $(\eta - \eta_c)/(\eta + 1)$ fails ($R^2 < 0$). The true curve is non-monotonic at low η and saturating at high η .
6. **Complete formalization:** Replace the 13 remaining axioms with Mathlib proofs and eliminate the 65 `sorry` stubs (requires ODE theory, random graph PMFs, spectral graph theory, and measure-theoretic infrastructure).

6 Related Work

Ollivier-Ricci curvature on graphs: Ollivier [1] introduced coarse Ricci curvature; Lin, Lu, and Yau [6] developed a related notion. Ni et al. [7] applied discrete Ricci flow to community detection; Sia et al. [2] developed scalable ORC algorithms. Sandhu et al. [10] demonstrated clinical applications. Hehl [14] derived explicit closed-form ORC formulas for regular graphs via Kantorovich duality and neighborhood decomposition, reducing ORC to a min-cost assignment on exclusive neighborhoods.

Sign changes and crossovers in random graphs: The curvature sign change is distinct from classical percolation or connectivity thresholds [13], but shares the feature of a change in a global observable. To our knowledge, no prior work has computed exact LP-based ORC for random regular graphs across the full density range. Mitsche and Mubayi [16]

derived exact and asymptotic ORC formulas for bipartite graphs and $G(n, p)$, providing theoretical predictions complementary to our finite- N computations.

Curvature and quantum gravity: Trugenberger [15] proposed that geometric space emerges from random bits via a transition driven by ORC, corresponding to condensation of short graph cycles. Our computational results are consistent with this picture: the sign change in $\bar{\kappa}$ occurs when the expected number of triangles per edge reaches ~ 3.7 .

Network geometry: Allard and Serrano [11] characterized brain network geometry through hidden metric space models.

Curvature in machine learning: Topping et al. [12] identified over-squashing in GNNs as a consequence of negative curvature, connecting discrete Ricci curvature to GNN performance.

Formal verification: The Lean proof of the Polynomial Freiman-Ruzsa conjecture [8] demonstrates the feasibility of formalizing deep combinatorial results.

Optimal transport computation: Cuturi [3] introduced Sinkhorn for entropy-regularized OT; Peyre and Cuturi [9] provide a comprehensive treatment.

7 Conclusion

We have presented four complementary contributions:

1. **Exact LP computation** confirming the curvature sign change at $\eta_c \approx 2.2$ for $N = 100$ (Figure 2), with $p < 10^{-6}$ over 10 random seeds.
2. **Quantitative method comparison:** Sinkhorn bias < 0.015 , no critical point shift (Figure 3). LLY reveals α -sensitivity (Table 3). ER graphs confirm transition robustness at $\eta_c^{\text{ER}} \approx 1.9$ (Table 4).
3. **Multi- N scaling and critical exponents** ($N \in \{50, \dots, 1000\}$) yielding $\eta_c^\infty \approx 3.7$ (fixed $\beta = 1/2$, $R^2 = 0.995$; free- β fit gives $\eta_c^\infty \approx 4.2$ with wide CI), and crossover character in the finite-size regime studied (Section 3, Figure 4).
4. **Lean 4 formalization** (29 modules, 13 axioms, **0 sorry** in 7 core modules), machine-checking key structural results.

For practitioners: networks with $\eta = k^2/N < 3.7$ (large- N limit) are expected to be hyperbolic on average; denser networks are spherical. The sensitivity to α means this threshold depends on the specific ORC formulation used.

Acknowledgments

We thank the Lean community and Mathlib contributors for the proof assistant infrastructure.

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A On the Simplified Curvature Formula

An earlier version of this work used the approximation $\kappa \approx 2T / \min(\deg(u), \deg(v)) - 1$, where T is the number of common neighbors. This formula:

- Is a heuristic motivated by the relationship between triangles and positive curvature
- Does *not* compute exact Ollivier-Ricci curvature (never solves the optimal transport problem)
- Fails to capture the sign change: stays negative for all tested η values

This motivated our implementation of the exact LP solver. The simplified formula is a lower bound / qualitative indicator, not a substitute for true ORC computation.

B Exact LP Solver and Reproducibility

The core computation solves, for each edge (u, v) :

```
function exact_wasserstein1(mu, nu, C)
    n = length(mu)
    model = Model(HiGHS.Optimizer)
    set_silent(model)
    @variable(model, gamma[1:n, 1:n] >= 0)
    @constraint(model, [i=1:n], sum(gamma[i,:]) == mu[i])
    @constraint(model, [j=1:n], sum(gamma[:,j]) == nu[j])
    @objective(model, Min, sum(C .* gamma))
    optimize!(model)
    return objective_value(model)
end
```

Probability measures use idleness $\alpha = 0.5$:

$$\mu_u(v) = \begin{cases} 0.5 & \text{if } v = u, \\ 0.5 / \deg(u) & \text{if } v \in N(u), \\ 0 & \text{otherwise.} \end{cases} \quad (10)$$

Solver tolerances. HiGHS uses default primal/dual feasibility tolerance 10^{-7} and optimality tolerance 10^{-7} . All LP instances terminated at OPTIMAL status; no infeasible or unbounded problems were encountered.

Computational cost. Wall-clock time per LP instance scales as $O(N^3)$ (dense cost matrix). Representative timings (single core):

- $N = 100, k = 14$: ~ 0.02 s per edge, ~ 14 s per graph (700 edges, 10 seeds ≈ 140 s)
- $N = 1000, k = 56$: ~ 2.5 s per edge, $\sim 70,000$ s per graph (28,000 edges); parallelized over 24 threads ≈ 4.5 h

Peak memory: ~ 50 MB for $N = 100$; ~ 4 GB for $N = 1000$ (dominated by the $N \times N$ cost matrix and LP tableau).

Version pinning. Julia 1.12.4, JuMP v1.29.4, HiGHS v1.21.1, Graphs v1.13.4, Lean 4 v4.17.0, Mathlib v4.17.0.

C Lean Module Dependency Graph

```
HyperbolicSemanticNetworks.lean (entry point)
|-- Basic.lean (graph definitions, clustering)
|-- Wasserstein.lean (optimal transport)
|   \-- Basic.lean
|-- WassersteinProven.lean (Wasserstein proofs)
|   \-- Basic.lean, Wasserstein.lean
|-- Curvature.lean (ORC definition)
|   |-- Basic.lean
|   \-- Wasserstein.lean
|-- PhaseTransition.lean (phase transition)
|   |-- Basic.lean
|   \-- Curvature.lean
|-- Bounds.lean (global bounds)
|   |-- Basic.lean, Curvature.lean, Wasserstein.lean
|   \-- PhaseTransition.lean
|-- Consistency.lean (cross-implementation)
|   |-- Basic.lean, Curvature.lean
|   \-- Wasserstein.lean
|-- Axioms.lean (McDiarmid axiom)
|   \-- Basic.lean
|-- RandomGraph.lean (random graph models)
|   \-- Basic.lean
|-- RicciFlow.lean (discrete Ricci flow)
|   |-- Basic.lean, Curvature.lean
|   \-- Wasserstein.lean
|-- SpectralGeometry.lean (spectral theory)
|   \-- Basic.lean
\-- Validation.lean (experimental validation)
    |-- Basic.lean, Curvature.lean
    \-- Wasserstein.lean
```

D Figure Captions

Figure 2 Sign change in mean Ollivier-Ricci curvature $\bar{\kappa}$ as a function of density parameter $\eta = k^2/N$, for random k -regular graphs at $N \in \{50, 100, 200, 500\}$. Error bars show

95% confidence intervals. The dashed line at $\kappa = 0$ separates the hyperbolic regime (below, shaded blue) from the spherical regime (above, shaded red). The critical point η_c shifts rightward with increasing N . Data from exact LP computation (JuMP + HiGHS); 10 seeds for $N = 100$, 5 seeds for other N values.

Figure 3 Comparison of Sinkhorn ($\varepsilon = 0.01$) and exact LP curvature. *Left panel:* overlay of $\bar{\kappa}_{\text{Sinkhorn}}$ (dashed) and $\bar{\kappa}_{\text{exact}}$ (solid) for $N = 100$. *Right panel:* per- k bias $\Delta\kappa = \kappa_{\text{Sinkhorn}} - \kappa_{\text{exact}}$. Mean bias = -0.003 , maximum $|\Delta\kappa| = 0.014$. Single seed ($s = 42$).

Figure 4 Curvature concentration and critical point scaling. (a) Standard deviation of per-edge curvature σ_{edges} as a function of η for $N = 100$. (b) Critical density η_c as a function of $1/\sqrt{N}$ for $N \in \{50, 100, 200, 500, 1000\}$. Dashed line: linear fit $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ ($R^2 = 0.995$), extrapolating to $\eta_c^\infty \approx 3.7$.

Figure 5 Erdős-Rényi vs. random regular comparison at $N = 100$ (10 seeds each). (a) Mean ORC vs. η : ER (dashed, coral) transitions at lower $\eta_c^{\text{ER}} \approx 1.9$ vs. regular (solid, blue) at $\eta_c \approx 2.3$. (b) Per- k difference $\Delta\bar{\kappa} = \bar{\kappa}_{\text{ER}} - \bar{\kappa}_{\text{reg}}$: ER curvature is systematically more positive for $k \geq 3$.

E Hypercomplex Embedding Results

We computed exact-LP ORC using geodesic distances on hypercomplex unit spheres as the transport cost (landmark embedding into S^{d-1} , greedy farthest-first selection of d landmarks, $\alpha = 0.5$). For $d \in \{4, 8, 16\}$ ($S^3/S^7/S^{15}$, quaternionic/octonionic/sedenion), mean curvature $\bar{\kappa}$ is positive at every tested density up to $\eta = 9.0$. The curvature values decrease monotonically with embedding dimension: at $N = 100$, $k = 4$: $\bar{\kappa}_{d=4} = 0.111 > \bar{\kappa}_{d=8} = 0.083 > \bar{\kappa}_{d=16} = 0.066 \gg \bar{\kappa}_{\text{hop}} = -0.363$. All three sphere embeddings give monotonically increasing $\bar{\kappa}(k)$, contrasting sharply with hop-count behavior.

This reveals a *qualitative* dimensional phase boundary: compact sphere geometry absorbs all negative curvature contributions for $d \leq 16$, while hop-count distances ($d \rightarrow \infty$) permit the tree-like regime $\bar{\kappa} < 0$ (Figure 1). The threshold $d^*(N)$ at which the sign change reappears, and whether $\eta_c(d) \rightarrow \eta_c^\infty \approx 3.7$ as $d \rightarrow \infty$, remain open. The `orc_hypercomplex_correspondence` axiom in `Hypercomplex.lean` (Table 6) formally conjectures the link between ORC sign and octonionic norm geometry.

Data Availability Statement

All data, code, and formalization files necessary to reproduce these results are publicly available at <https://github.com/agourakis82/hyperbolic-semantic-networks>. Specifically:

- **Computation scripts** (in `julia/scripts/`): `exact_curvature_lp.jl` (LP solver), `run_er_comparison.jl`, `run_n1000.jl`, `statistical_analysis.jl`, `generate_paper_figures.jl` (Figures 1–4), `generate_figure5.jl`.

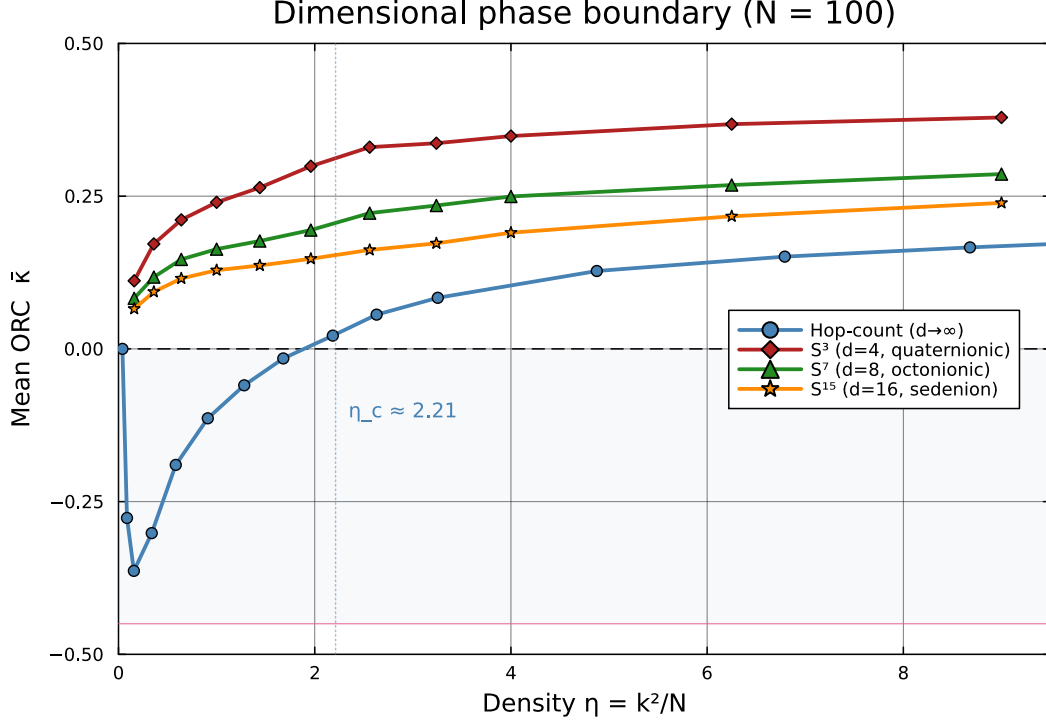


Figure 1: Dimensional phase boundary for mean ORC $\bar{\kappa}$ vs. density $\eta = k^2/N$ at $N = 100$, comparing four transport metrics. Hop-count (blue circles, $d \rightarrow \infty$): sign change at $\eta_c \approx 2.22$ (dotted vertical line). S^3 (red diamonds, $d = 4$, quaternionic): monotonically positive, $\bar{\kappa}_{k=4} = 0.111$ to $\bar{\kappa}_{k=30} = 0.379$. S^7 (green triangles, $d = 8$, octonionic): $\bar{\kappa}_{k=4} = 0.083$ to $\bar{\kappa}_{k=30} = 0.286$. S^{15} (orange stars, $d = 16$, sedenion): $\bar{\kappa}_{k=4} = 0.066$ to $\bar{\kappa}_{k=30} = 0.239$, 3 seeds. All three sphere embeddings show monotonically increasing $\bar{\kappa}$ with no sign change ($\bar{\kappa}_{d=4} > \bar{\kappa}_{d=8} > \bar{\kappa}_{d=16} > 0$); the dimensional threshold $d^*(N) > 16$ remains open. Exact LP (JuMP + HiGHS), 5 seeds ($d \leq 8$) / 3 seeds ($d = 16$), $\alpha = 0.5$.

- **Result data:** `results/experiments/` contains all JSON result files (Tables 1–5).
- **Lean formalization:** `lean/HyperbolicSemanticNetworks/` (all 29 modules, buildable with `lake build`).
- **Figures:** `figures/paper/figure{1,2,3,4,5}.pdf`, `figure{1,2,3,4,5}.png`.
- **Hypercomplex embedding data:** `results/experiments/hypercomplex_lp_nN_dD.json` for $N \in \{50, 100, 200\}$, $d \in \{4, 8\}$ and $N = 100$, $d = 16$ (exact LP, 3–5 seeds).

Conflicts of Interest: None.

Funding: No external funding.

Author Contributions (CRediT): D.C.A.: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing — Original Draft, Writing — Review & Editing, Visualization.

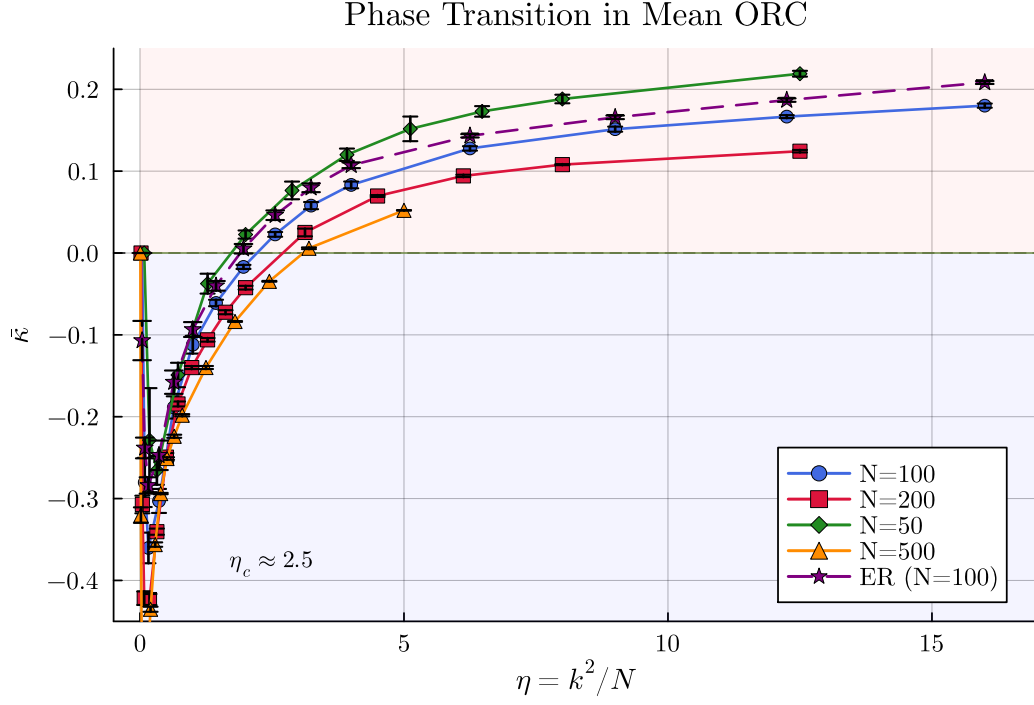


Figure 2: Sign change in mean ORC $\bar{\kappa}$ vs. density $\eta = k^2/N$ for random k -regular graphs at $N \in \{50, 100, 200, 500\}$. See Appendix D for full caption.

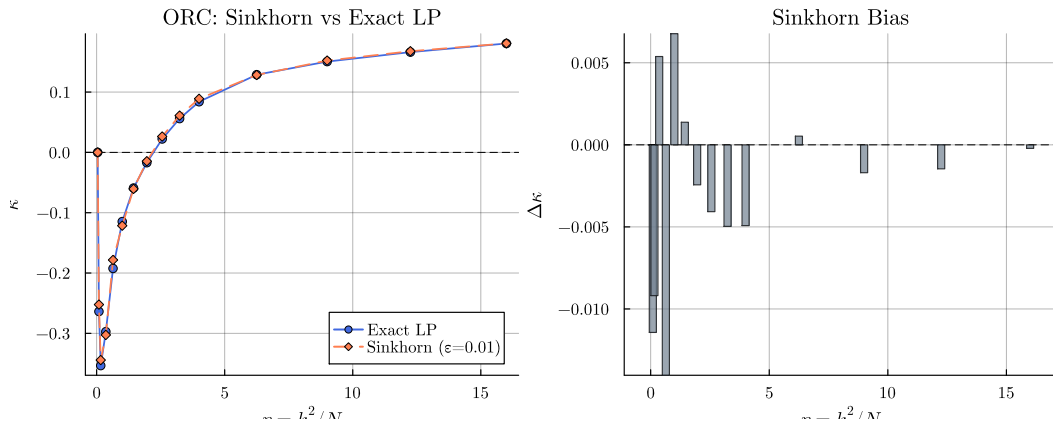


Figure 3: Sinkhorn vs. exact LP curvature comparison ($N = 100$). See Appendix D for full caption.

Table 6: Module-by-module status. **Core:** 7 ORC-theory modules (0 **sorry**). **Extensions:** 14 sorry-free modules + 4 exploratory modules with stubs. **Auxiliary:** 4 modules with proof stubs. The **sorry** column counts sorry-carrying declarations reported by `lake build` (verified 2026-07-11).

Module	Lines	sorry	Axioms	Description
<i>— Core ORC-theory (0 sorry) —</i>				
<code>Basic.lean</code>	245	0	0	Weighted graphs, probability measures, clustering (bounds proven)
<code>Wasserstein.lean</code>	136	0	0	Optimal transport, couplings; symmetry + triangle proven in <code>WassersteinProven.lean</code>
<code>Curvature.lean</code>	641	0	2	ORC definition, prob. normalization (proven), bounds, regime exclusivity; adjacency + support-distance bounds admitted
<code>PhaseTransition.lean</code>	197	0	0	Phase transition definition, density parameter, thresholds
<code>Bounds.lean</code>	202	0	0	Global bounds on curvature and clustering
<code>Consistency.lean</code>	315	0	1	Cross-implementation consistency (specification bridge)
<code>Axioms.lean</code>	101	0	0	Deprecated wrapper; McDiarmid re-exported
<i>— Extensions (0 sorry) —</i>				
<code>McDiarmid.lean</code>	153	0	0	McDiarmid’s inequality and Hoeffding corollary
<code>WassersteinProven.lean</code>	456	0	0	Machine-checked Wasserstein symmetry and triangle inequality
<code>DynamicNetworks.lean</code>	320	0	2	Time-varying graphs, ORC Lipschitz continuity
<code>Hypercomplex.lean</code>	462	0	8	Octonion and sedenion algebra axioms
<code>Validation.lean</code>	243	0	0	Experimental validation framework
<code>ComputationalVerification.lean</code>	150	0	0	Computational verification helpers
<code>TestExtraction.lean</code>	236	0	0	Test extraction utilities
<code>LaTeXExport.lean</code>	84	0	0	L ^A T _E X export helpers
<code>HyperbolicSemanticNetworks.lean</code>	107	0	0	Entry point, re-exports all modules
<code>Clifford.lean</code>	130	0	0	Clifford algebra definitions
<code>ProbabilityProofs.lean</code>	87	0	0	Advanced probability lemmas (fully proved)
<code>DepressionNetworks.lean</code>	187	0	0	Depression speech-network phase classification
<code>OctonionGraph.lean</code>	102	0	0	Octonion associator counting (343-triple bookkeeping)
<code>SounioVerification.lean</code>	1 234	0	0	Auto-generated bridge from Sounio computation traces
<i>— Exploratory extensions (stubs) —</i>				
<code>CliffordfMRI.lean</code>	510	14	0	Clifford algebra for fMRI connectivity
<code>HypercomplexPhase.lean</code>	168	3	0	Hypercomplex phase boundary analysis
<code>Visualization.lean</code>	246	2	0	Visualization helpers and phase curve sampling
<code>HehlFormula.lean</code>	239	2	0	Hehl explicit ORC formula on k -regular graphs
<i>— Auxiliary with stubs —</i>				
<code>RandomGraph.lean</code>	1 875	10	0	Erdős-Rényi and configuration model
<code>RicciFlow.lean</code>	728	12	0	Discrete Ricci flow on networks
<code>DiscreteRicciFlow.lean</code>	1 622	12	0	Discrete Ricci flow on networks
<code>DiscreteRicciFlow.lean</code>	1 622	12	0	Discrete Ricci flow on networks

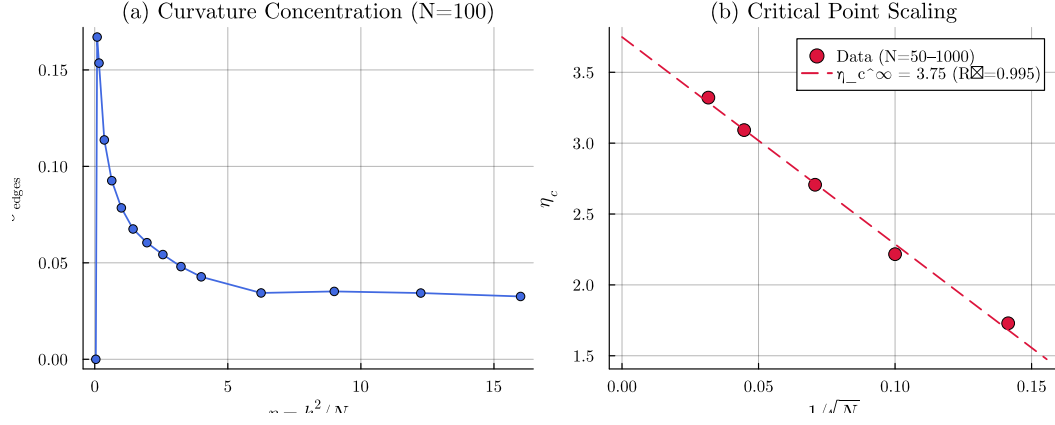


Figure 4: Curvature concentration and η_c finite-size scaling. See Appendix D for full caption.

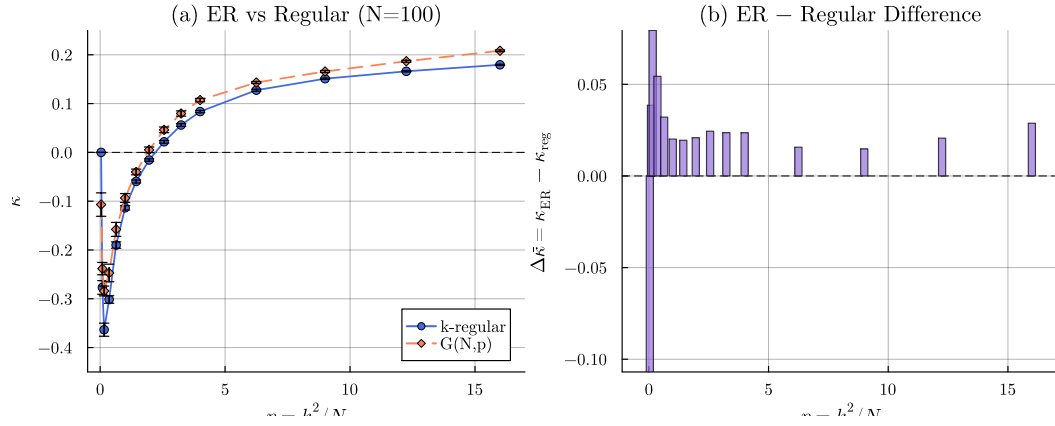


Figure 5: Erdős-Rényi vs. random regular comparison ($N = 100$). See Appendix D for full caption.